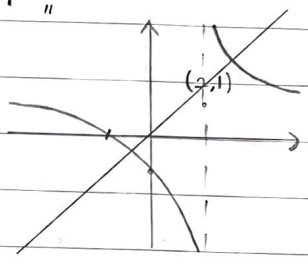
6.1 HW #1(2) #3(2) #4(2) #6(2) #9 //

#1 (2)

$$f(x) = \frac{x+2}{x-2} \quad [0, 2) \qquad \frac{x+2}{x-2} = 1 + \frac{4}{x-2}$$



$$f'(x) = \frac{(x-2)-(x+2)}{(x-2)^2} = \frac{-4}{(x-2)^2} < 0$$

∴ f(x)는 역함수 존재 //

$f(0) = -1, \quad f(2) = \infty$

∴ 역함수의 정의역 : $x \leq -1$ //

#3 (2) $(f^{-1})'(-2) = ?$

$$f(x) = 3x - \frac{1}{x^3}, \quad c = -2$$

$$3x - \frac{1}{x^3} = -2$$

$$x = -1$$

$$(f^{-1})'(-2) = \frac{1}{f'(-1)} = \frac{1}{6}$$

$$f'(x) = 3 + \frac{3x^2}{x^6} = 3 + \frac{3}{x^4}$$

∴ $\frac{1}{6}$ ▨

#4 (2)

$$f(x) = \frac{1+3x}{5-2x} \; ; \; I = \left(\frac{5}{2}, \infty\right)$$

$$x = \frac{1+3y}{5-2y}$$

$$5x - 2xy = 1 + 3y$$

$$(2x+3)y = 5x - 1$$

$$y = \frac{5x-1}{2x+3} \qquad \therefore f'(x) = \frac{5x-1}{2x+3}$$ //